

Insect-Inspired Tactile Contour Sampling Using Vibration-Based Robotic Antennae

Thierry Hoinville¹, Nalin Harischandra¹, André F. Krause^{1,2}, and Volker Dürr¹

¹ Department of Biological Cybernetics, University of Bielefeld, Germany

² Cognitive Interaction Technology – Center of Excellence (CITEC),

University of Bielefeld, Germany

`thierry.hoinville@uni-bielefeld.de`

Abstract. Compared to vision, active tactile sensing enables animals and robots to perform unambiguous object localization, segmentation and shape recognition. Recently, we proposed a bio-inspired, CPG-based, active antennal control model, so-called Contour-net, which captures essential characteristics of antennal behavior in climbing stick insects. In simulation, this model provides a robust and effective way to trace contours and classify various 3D shapes. Here, we propose a physical robotic implementation of Contour-net using vibration-based active antennae. We show that combining tactile contour tracing with vibration-based distance estimation yields fairly accurate localization of contact events in 3D space.

Keywords: Tactile contour tracing, Tactile sampling, Tactile localization, Antenna, Hopf oscillator, Stick insect.

1 Introduction

Tactile sensing enables humans and animals to actively perceive extensive properties of surrounding objects that are hard, if not impossible, to obtain through vision [10,18,8,13]. Notable examples are surface texture, chemical properties, temperature, compliance and humidity. However, even regarding properties that are known to be routinely processed from visual input, like object contour and distance, direct physical contact yields unambiguous robust estimates. In fact, the sense of touch is independent of light conditions and directly provides reliable three-dimensional information useful for object localization, segmentation and shape recognition.

Insect antennae and mammal whiskers have inspired robotic research in the area of tactile sensors. Early work by Kaneko et al. [3] describes an artificial antenna using a flexible beam capable of detecting 3D contact locations and surface properties. Russel and Wijaya [15] applied an array of whiskers that passively scan over an object to recognize its shape using advanced pre-processing contact-points and decision trees. Robotic whisker arrays were used by Solomon and Hartmann [17] to generate 3D spatial representations of the environment and extract object shapes. Related work done by Kim and Möller [4] used a vertical whisker array to detect the vertical shape and curvature of objects. Sullivan

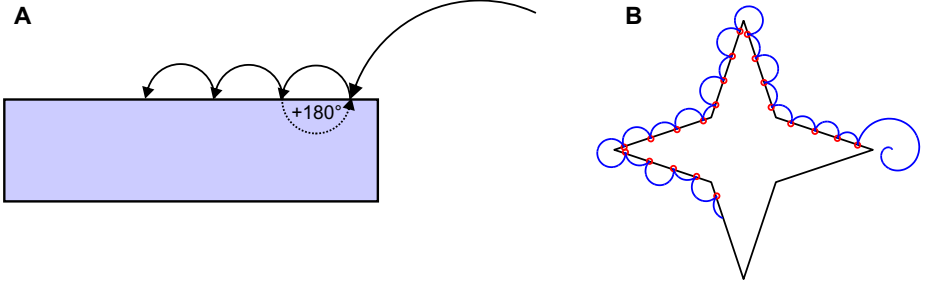


Fig. 1. **A:** Tactile contour-tracing template. Each contact induces a 180° phase forwarding of a circular movement. Combined with velocity control of a tactile probe, the resulting change in movement velocity and direction causes the probe to “bounce off” the surface at every contact event, resulting in a successive scan along the object’s shape. **B:** 2D simulation of contact-triggered contour tracing. *Black:* star-shaped object. *Blue:* trajectory of the antennal tip. *Red dots:* contact locations.

et al. [19] performed Bayesian classification of surface texture using an array of actuated hair-like elements. Drawing inspiration from the American cockroach, Lee et al. [9] developed an artificial antenna and a control model for high-speed wall following of a wheeled robot. Recently, Mongeau et al. [11] added hairs to this antenna and studied implications for mechanics and sensing.

Recently, we proposed a bio-inspired, CPG-based, active antennal control model called “Contour-net” [6]. This template model [9] captures essential characteristics of antennal behavior in walking and climbing stick insects: rhythmic searching movements using an antenna with two revolute joints with strong coupling and a contact-triggered switch to local sampling [16,5,7]. Specifically, the antennal velocity is driven by a CPG modulated by a single, binary sensor signal. Upon the first contact with an obstacle (Fig. 1A), the behavior switches from a broad antennal searching pattern to a local contour sampling pattern with higher frequency and lower amplitude. Each contact event triggers a 180° phase shift in the CPG, which leads the antenna to “bounce off” the surface of the object repeatedly. In [6], we have shown in simulation that this simple model provides a robust and effective way to trace contours of 2D and 3D objects of various shapes (Fig. 1B). Furthermore, sampled contact points may directly be used for neural network-based shape classification [6].

In this paper, we extend the previous work by implementing Contour-net on a physical robotic platform equipped with two vibration-based active antennae as developed in [12]. To assess the effectiveness of our implementation, we report its performance relating to 3D localization.